

Nicholas Chin Bo You

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Education

University of Utah
Salt Lake City, UT

August 2024 – Present

BS in Physics with Computational Physics and Astronomy Emphasis | BS in Applied Mathematics
CGPA: 3.872 | Dean's List: Fall 2024–Summer 2026

Research Experience

Research Assistant

Fall 2024 – Present

University of Utah Particle Astrophysics Research Group

- Investigating extended gamma-ray emission from the Geminga pulsar using very-high-energy gamma-ray observations.
- Reproduced published VERITAS results for the Crab Nebula and Cassiopeia A using **Gammapy** analysis workflows.
- Analyze stellar intensity interferometry observations of β Canis Majoris to search for photospheric pulsation signatures.
- Advisor: Prof. David Kieda, Prof. Tugdual LeBohec

Quantum State Preparation Algorithms for Astronomical Images

Spring 2026 – Present

Independent Research

- Investigating efficient quantum state-preparation algorithms for astronomical image encoding by exploiting image sparsity and nonzero structure to reduce quantum circuit complexity.
- Advisor: Prof. Massood Tabib-Azar

Shared Risk-Capital Constraints in Dynamic Option Dealer Hedging

Manuscript submitted

Collaborative Research

- Formulated dynamic option-dealer hedging under shared funding limits as a shared-constraint generalized Nash equilibrium and analyzed its variational equilibrium characterization.

Publications & Selected Presentations

A. Archer, J. P. Aufdenberg, ..., **N. B. Y. Chin**, ..., The VERITAS Collaboration. "Measurement of the photosphere oblateness of γ Cassiopeiae via Stellar Intensity Interferometry with the VERITAS Observatory." *The Astrophysical Journal* **995**, 191 (2025).

University of Utah 2026 Undergraduate Summer Symposia - [Poster Presentation](#) 

2026 Spring Quantum Computing Focused Reading Group (NSF Research Training Group Affiliated)

APS Four Corners Section 2025 Annual Meeting - [Poster Presentation](#) 

Skills

- Languages: English, Mandarin Chinese, Cantonese Chinese, Hakka Chinese, Bahasa Melayu
- Programming Languages: Python, C/C++, Rust, Java, Bash, $\text{\LaTeX}$
- Scientific Computing: CERN ROOT, Gammapy, Qiskit, Jupyter Notebook, NumPy, Matplotlib, MATLAB
- Operating Systems: Linux / Unix

Selected Honors & Awards

Summer Undergraduate Research Fellowship (SURF), 2026 | Undergraduate Research Opportunity Program (UROP), 2026 | SLMATH-SACNAS Travel Award, 2026 | Thomas J. Parmley Scholarship, 2026 | Talent Scholarship, 2025 | MIT iQuHACK 2026 – BlueQubit Problem: World Rank 35, 2nd Highest Score

Leadership Experience

Lead Organizer - Qiskit Fall Fest 2026

Salt Lake City, UT

IBM Quantum

Aug 2026 – Present

- Selected by IBM Quantum to lead the organization of a Qiskit Fall Fest event at the University of Utah.

Chair - Undergraduate Student Advisory Committee

U of U

U of U Physics Department

Oct 2025 – Present

- Organized department-wide events and led a 40+ student town hall to communicate undergraduate feedback to faculty leadership.